Recognizable extensions of colored graphs

Human Compression Project

Abstract

A construction can carry every symmetry of its input and still acquire additional symmetries. We separate these two questions by a restriction and extension criterion for permutation groups and their distinguished subgroups. Intrinsic recognition of retained vertices, rigidity over those vertices, and explicit extension identify the full group. Component boundaries provide a useful local rigidity test. We give complete proofs for the chamber completion and cleaning constructions of Díaz Ramos, Molinier and Viruel. Their full ordered-pair encoding and finite normal-pair representation proof are shared here by both applications. The first application records those source statements and recovers the recursive symmetric-group family. A second application treats the constructions of Leemans, Stokes and Tranchida. We include the full kernel created by repeated shadows, and the free-basis coset geometry and normalizer arguments needed by its infinite examples. Distinct-length arms preserve ordered attachments, while a thin-doubling construction separates the connected-chamber splitting theorem from the unqualified statement.

All sections before the free-basis construction concern finite incidence systems. The free-basis sections explicitly allow infinitely many objects with a finite palette; the finite convention in the opening definitions is local to the finite constructions. The thin-doubling proposition returns to finite systems.

1 Restriction, extension and component boundaries

An incidence system in this paper is a nonempty finite simple graph X, together with a surjective proper coloring $t : V(X) \to I$. Thus adjacent vertices have different colors. A *correlation* is a graph automorphism f that permutes the color classes: there is a permutation σ_f of I such that $t(f(v)) = \sigma_f(t(v))$ for all v. Surjectivity makes σ_f unique. Write $C(X)$ for the correlation group and $K(X)$ for its subgroup preserving each color. The map $f \mapsto \sigma_f$ is a homomorphism, so $K(X)$ is normal and $C(X)/K(X)$ embeds in $\mathrm{Sym}(I)$. In general $C(X)$ need not be the full uncolored graph automorphism group.

A *chamber* is a clique meeting every color class. An incidence system is an *incidence geometry* when each clique, including the empty clique, is contained in a chamber. No transitivity, thinness or residual connectedness is part of this definition. These are the conventions of Díaz Ramos–Molinier–Viruel [1], Definitions 2.1–2.3.

The following elementary fact is the common group-theoretic argument. It is stated for arbitrary permutation groups; finiteness enters the graph constructions later.

Theorem 1 (Restriction and extension)**.** *Let $\iota : V \hookrightarrow W$ be an injection, let $A \leq \mathrm{Sym}(V)$ and $B \leq \mathrm{Sym}(W)$, and let $A_0 \leq A$ and $B_0 \leq B$. Suppose that:*

1. *for each $b \in B$ there is $a \in A$ such that $b\iota = \iota a$;*
2. *a member of B fixing $\iota(V)$ pointwise is the identity;*
3. *every $a \in A$ has an extension $b \in B$ satisfying $b\iota = \iota a$;*
4. *for such a pair (b, a), one has $b \in B_0$ if and only if $a \in A_0$.*

Then restriction induces an isomorphism $B \cong A$ carrying B_0 onto A_0. Every extension in the third condition is unique, and the resulting extension map is the inverse homomorphism.

Proof. The injection ι makes the element a in the first condition unique; write $\rho(b) = a$. For $b, c \in B$,

$$bc\iota = b\iota\rho(c) = \iota\rho(b)\rho(c),$$

so uniqueness gives $\rho(bc) = \rho(b)\rho(c)$. The same argument gives $\rho(1) = 1$. Its kernel consists exactly of the elements fixing $\iota(V)$ pointwise, and is trivial by the second condition. The third condition makes ρ surjective. The last condition gives $\rho(B_0) = A_0$: containment follows immediately, while a preimage of any $a \in A_0$ belongs to B_0. Thus ρ is the asserted pair isomorphism. Injectivity gives uniqueness of extensions, and the inverse of an isomorphism is a homomorphism. □

For colored graphs X, Y, one applies this theorem to $A = C(X)$, $B = C(Y)$ and their palette kernels. Recognizing a retained copy of $V(X)$ is not enough: the restricted permutation must preserve the *reconstructed* old incidences and colors. In particular, old edges need not remain edges of Y. Conversely, an explicit action of $C(X)$ on Y establishes only the extension condition, until rigidity rules out further correlations.

Here is a local criterion for that rigidity. All induced graphs and boundaries in its statement are taken in the constructed graph Y.

Lemma 2 (Recognition by component boundaries)**.** *Let $S \subseteq V(Y)$ be invariant under a group B of correlations of Y. For each connected component D of $Y \setminus S$, define its boundary*

$$\partial D = \{s \in S : s \text{ has a neighbor in } D\}.$$

Assume distinct components have distinct boundaries. Suppose also that any correlation of the induced colored graph on $D \cup \partial D$ fixing ∂D pointwise fixes D pointwise. Then a member of B fixing S pointwise is the identity on Y.

Proof. Such a member permutes the components of $Y \setminus S$ and sends each boundary to itself. Distinctness of the boundary sets therefore fixes each component setwise. Its restriction to $D \cup \partial D$ is a correlation, fixes the boundary pointwise, and is the identity by the local assumption. This holds for every component; it is already the identity on S. □

Only color classes occurring in an induced subgraph are used when speaking of its restricted coloring. A global correlation preserving that subgraph does permute those classes, even if it permutes other colors outside the subgraph. Thus the local hypothesis in Lemma 2 does not silently replace correlations by color-preserving automorphisms.

2 Completing edges to chambers

The following construction and theorem are due to Díaz Ramos, Molinier and Viruel [1, Theorem 4.1]. We retain their construction and prove all four conditions of Theorem 1.

Theorem 3 (Chamber completion)**.** *Let X be a finite properly colored triangle-free graph of minimum degree at least two, with palette I of size r. For every edge $e = \{v, w\}$ and every $i \in I \setminus \{t(v), t(w)\}$, introduce a vertex $u_{e,i}$ of color i. Make*

$$B_e = \{v, w\} \cup \{u_{e,i} : i \notin \{t(v), t(w)\}\}$$

a complete graph, retaining all old vertices and adding no other edges. The resulting graph $J(X)$ is an incidence geometry on the same palette and restriction gives

$$(C(J(X)), K(J(X))) \cong (C(X), K(X)).$$

It has $|V(X)| + (r-2)|E(X)|$ vertices and the same connected components as X.

Proof. Every B_e has exactly one vertex of each color. A clique containing a new vertex lies entirely in its B_e, since that vertex has no neighbors outside B_e. A clique consisting of old vertices has size at most two by triangle-freeness. A two-vertex clique supplies its edge e; a singleton extends using an incident edge; the empty clique extends because X has an edge. Hence every clique is contained in a chamber.

A new vertex has degree $r-1$. At an old vertex v, the sets $B_e \setminus \{v\}$, for edges e through v, are pairwise disjoint. Consequently

$$\deg_{J(X)}(v) = \deg_X(v)(r-1). \tag{1}$$

The hypotheses imply $r \geq 2$ and $\deg_X(v) \geq 2$, so the old vertices are exactly those of degree greater than $r-1$. Every correlation recognizes them. Their induced graph is X, and their old colors are unchanged; the restriction therefore belongs to $C(X)$.

Suppose a correlation fixes every old vertex. Every old color occurs, so it fixes every color. If a new vertex has old neighbor set $\{v, w\}$ and color i, it must be $u_{\{v,w\},i}$. Thus all new vertices are fixed and restriction is injective. If $f \in C(X)$ induces $\sigma_f \in \mathrm{Sym}(I)$, define

$$\widetilde{f}(v) = f(v), \qquad \widetilde{f}(u_{e,i}) = u_{f(e),\sigma_f(i)}.$$

This is a bijection, with inverse defined from f^{-1}. It carries each B_e onto $B_{f(e)}$, and hence preserves adjacency in both directions. It fixes all colors precisely when f does. Theorem 1 now identifies the pairs. When $r = 2$ no vertices are added, and this proof still applies.

There are $r-2$ new vertices per edge, giving the count. Every new vertex is attached within the component of its edge, and all old edges are retained. No components are joined or lost. □

The source's degree argument on physical page 7 uses r in place of $r-1$. Equation (1) is the actual degree calculation. The strict separation needed for its theorem is unaffected. The statement concerns this construction and these hypotheses; it does not assert that they are logically necessary for every conceivable pair-preserving completion.

3 Cleaning arbitrary incidence systems

The next construction is [1, Theorem 4.2]. We include the rigidity of its tail and the isolated-vertex case, since those are the local obligations that make its symmetry assertion valid.

Theorem 4 (Cleaning)**.** *Every nonempty finite properly colored graph X has a finite properly colored, triangle-free graph $L(X)$ of minimum degree at least two satisfying*

$$(C(L(X)), K(L(X))) \cong (C(X), K(X)).$$

It preserves the connected-component decomposition and adds exactly $q(2M+5)$ vertices, where

$$M = \max_{i \in I} |t^{-1}(i)|, \qquad W = \{v : \deg_X(v) \leq 1\}, \qquad q = |E(X)| + |W|.$$

Its palette consists of the old colors and two new colors.

Proof. Since X is nonempty, $M \geq 1$; also $q \geq 1$, since otherwise a nonempty edgeless graph would have no vertices of degree at most one. For each edge e and each $w \in W$ take a separate gadget with vertices $u_0, \ldots, u_{2M+4}$, consecutive edges $u_j u_{j+1}$, and the two chords

$$u_{2M+4}u_{2M-1}, \qquad u_{2M+3}u_{2M}. \tag{2}$$

Delete all old edges. For $e = \{v, w\}$ join both endpoints to the root $u_{e,0}$ of its edge gadget. Join each $w \in W$ to $u_{w,0}$; if w was isolated, also join it to $u_{w,2}$. Retain old colors, color even gadget indices by α and odd indices by β, where α, β are distinct new colors.

Every new edge joins different colors. An old vertex of degree at least two retains that degree; a former leaf acquires a second neighbor; a former isolated vertex acquires its two displayed neighbors. Every gadget vertex has degree at least two. There are no old–old edges. The new induced graph is bipartite. A triangle containing an old vertex would require two adjacent new neighbors in one gadget, but the only old vertex with two neighbors in one gadget was isolated, and its neighbors u_0, u_2 are not adjacent. Thus $L(X)$ is triangle-free and has the required minimum degree.

The two new color classes have sizes $q(M+3)$ and $q(M+2)$. These sizes are distinct and both exceed M, the largest old class size. Every correlation therefore fixes each of these classes and preserves the old vertex set. Two distinct old vertices were adjacent in X exactly when they have a common new neighbor in $L(X)$: their edge gadget supplies its root, and every other gadget has only one old neighbor. This reconstructs the old graph as well as its color partition, so restriction belongs to $C(X)$.

To establish rigidity, delete all old vertices. The components are precisely the separate gadgets. Each edge gadget has its old endpoint pair as boundary; each additional gadget has its one old vertex as boundary. The graph is simple and at most one additional gadget is attached to a given old vertex, so all these boundaries are distinct.

Consider a correlation of a gadget together with its boundary that fixes the boundary. For an edge gadget or a former leaf, the root is the unique gadget vertex adjacent to the boundary. For an isolated vertex the two attached vertices u_0, u_2 have different degrees in this induced graph: the first has degree two and the second at least three. Hence the root is fixed in all cases. Starting at the root forces successive vertices until u_{2M-1}. At that vertex the forward neighbors u_{2M} and u_{2M+4} have induced-gadget degrees three and two respectively, so both are fixed. At u_{2M} the remaining neighbors u_{2M+1} and u_{2M+3} have induced-gadget degrees two and three respectively. Finally u_{2M+2} is forced. These branch degrees are taken within the gadget, after its component has been recognized; an isolated attachment therefore causes no exception when $M = 1$. Lemma 2 now gives pointwise rigidity over all old vertices.

A correlation f of X preserves degrees. Extend it by

$$u_{e,j} \longmapsto u_{f(e),j}, \qquad u_{w,j} \longmapsto u_{f(w),j}.$$

The complete edge list, including the isolated attachment, is preserved, and the inverse is the same construction from f^{-1}. Both new colors are fixed; old colors are fixed exactly when f fixes them. Thus Theorem 1 gives the asserted pair isomorphism. There are q gadgets of $2M+5$ vertices. Replacing an old edge preserves connectivity between its endpoints and does not join different old components; all other gadgets attach to one old vertex. This proves the component claim. □

Corollary 5. *Every finite incidence system has a finite incidence geometry with the same correlation group and the same distinguished type-preserving subgroup. Connected inputs give connected outputs.*

Proof. Apply Theorem 3 to $L(X)$ from Theorem 4. Both restriction maps identify the distinguished kernels. Each construction preserves the component decomposition. □

4 An explicit representation of a normal pair

We use the finite colored-graph definition of incidence geometry and the groups $C(X), K(X)$ from Section 1. Correlations permute color classes; $K(X)$ fixes them individually. For a finite group G and a normal subgroup H, the following is Theorem A of Díaz Ramos, Molinier and Viruel [1], with the rank and connectedness of their construction stated explicitly.

Theorem 6. *Every finite normal pair $H \trianglelefteq G$ is isomorphic to $(C(\Gamma), K(\Gamma))$ for a finite connected incidence geometry Γ. For $G \neq 1$ the construction below has rank $[G : H] + 3$; for $G = 1$ it has rank one. No fixed rank-two assertion is made.*

Normality is necessary because $K(\Gamma)$ is the kernel of the palette action. In particular G/H embeds in a symmetric group on the palette, so rank two would force $[G : H] \leq 2$. This elementary restriction is not evaded by allowing correlations that exchange types.

Theorem 7 (Ordered-pair encoding)**.** *Let $G \neq 1$ be finite and $H \trianglelefteq G$. There is a connected properly colored triangle-free graph $X(G,H)$ of minimum degree two with*

$$(C(X(G,H)), K(X(G,H))) \cong (G,H).$$

For $q = |G|$ it has $(q^3 + 9q^2 - 8q)/2$ vertices and $[G : H] + 3$ colors.

Proof. This is the explicit construction of [1, Theorem 3.1]. Enumerate $G = \{g_1 = 1, g_2, \dots, g_q\}$. Take a vertex P_x for each $x \in G$. Label the ordered pair (x, y), for $x \neq y$, by k when $yx^{-1} = g_k$. Thus $k \geq 2$. Introduce separate vertices

$$T_{x,y}, \quad S_{x,y,0}, \dots, S_{x,y,k+2}$$

for every ordered pair. Suppressing the pair subscripts within one gadget, add exactly the edges

$$P_xS_0, \quad S_0P_y, \quad S_iS_{i+1}\ (0 \leq i < k+2), \quad P_xT, \quad TS_{k+2}, \quad S_{k+2}P_y. \tag{3}$$

There are no edges between original vertices and no other edges. Color P_x by its coset xH. Give all T-vertices one new color, the even-indexed S-vertices a second, and the odd-indexed ones a third. All three colors occur and are disjoint from the coset palette.

Every displayed edge joins different colors: only consecutive chain vertices use the two parity colors together, and every other edge has distinct vertex kinds or joins the old and new palettes. A single gadget has no triangle. Its short paths between the old endpoints have lengths two and three, while the chain between S_0, S_{k+2} has length $k + 2 \geq 4$. More directly, no adjacent pair of internal neighbors of either old endpoint occurs in the edge list, and the internal chain with its one attached T has no triangle. Distinct gadgets meet only at old vertices, which are pairwise nonadjacent, so no triangle can use different gadgets. Every pair of old vertices is connected through its gadget and every new vertex connects to its endpoints. Thus the graph is connected.

The degrees of old vertices are $4(q-1)$, those of S_0, S_{k+2} are three, and all remaining vertices have degree two. Consequently the old vertices are exactly those of degree at least four, including when $q = 2$. Let an uncolored graph automorphism induce $P_x \mapsto P_{\rho(x)}$. The degree-two neighbors of P_x are precisely the $T_{x,z}$. Among them $T_{x,y}$ is the unique one at distance two from P_y: its other neighbor is the terminal vertex $S_{x,y,k+2}$, adjacent to exactly that old vertex P_y. The terminal vertex is also the unique degree-three neighbor of $T_{x,y}$.

Starting at this terminal vertex, the path with degree-two internal vertices that ends at a degree-three vertex is intrinsic. It is the chain ending at $S_{x,y,0}$ and has length $k + 2$. The other route through $T_{x,y}$ ends at an old vertex of degree at least four. Hence the graph recovers both the orientation and the full label k of each ordered pair. Once the old permutation is fixed, all gadget vertices are forced. Conversely a permutation of the old vertices preserving these labels extends uniquely by preserving every gadget index. This is exactly the recognition and extension mechanism of Theorem 1, applied first to the labeled ordered-pair structure.

Label preservation is the equation

$$\rho(y)\rho(x)^{-1} = yx^{-1} \qquad (x \neq y).$$

Taking $x = 1$ gives $\rho(y) = y\rho(1)$ for $y \neq 1$, and the same equality holds at $y = 1$. Thus the full uncolored automorphism group consists of right translations. For ordinary composition of permutations use the homomorphism

$$g : P_x \longmapsto P_{xg^{-1}},$$

with the same right translation on both indices of every gadget. The inverse in this formula fixes the left/right convention; it does not alter the graph construction.

All translations preserve the three auxiliary colors. Normality of H makes them permute the cosets xH, so every uncolored automorphism is a correlation. A translation fixes all coset colors if and only

if $g \in H$: necessity follows from the coset of the identity, and normality gives sufficiency on every coset. This proves the exact pair assertion.

For each label $k = 2, \dots, q$ there are q ordered pairs, with $k+4$ internal vertices per pair. The total is

$$q + q \sum_{k=2}^{q} (k+4) = \frac{q^3 + 9q^2 - 8q}{2}.$$

This is the size of the borrowed construction, not an optimality statement. □

Proof of Theorem 6. For $G \neq 1$, apply Theorem 3 to $X(G, H)$. It preserves the correlation pair, palette and connectedness, so the resulting geometry has rank $[G : H]+3$. For $G = 1$, use a single vertex with one color. Its correlation group and type-preserving subgroup are both trivial, and its only nonempty clique is a chamber. □

Theorem 5 supplies a separate conclusion: any nonempty finite incidence system, including one with isolated vertices or triangles, can first be cleaned and then completed without changing its pair. Cleaning preserves components rather than making a disconnected input connected. Thus the connectedness asserted above comes from the explicit encoder; it is not an unstated hypothesis or conclusion of the arbitrary-input case.

5 Repeated shadows and the restriction kernel

This section includes the elementary shadow-fiber theorem of the earlier phase196 manuscript. It makes explicit the rigidity condition in Theorem 1; the old objects being recognizable does not force the new objects to be determined by them.

Let P be a finite typed incidence system, J a finite set of new types, and Q_j pairwise disjoint finite sets, disjoint also from P. For $q \in Q_j$ prescribe its *complete old shadow* $s_j(q) \subseteq P$. Retain the old incidences, put $p * q$ exactly when $p \in s_j(q)$, and make all objects of distinct new types incident. There are no incidences within a new type. Write Γ for the extension and $m_j(S) = |s_j^{-1}(S)|$. Let

$$H = \{h \in K(P) : m_j(hS) = m_j(S) \text{ for every } j, S\}.$$

Absent shadows have multiplicity zero.

Theorem 8 (The complete shadow kernel)**.** *Restriction to P gives the split exact sequence*

$$1 \longrightarrow \prod_{j,S} \mathrm{Sym}(s_j^{-1}(S)) \longrightarrow K(\Gamma) \longrightarrow H \longrightarrow 1.$$

Consequently restriction is faithful exactly when every nonempty shadow fiber has one object. A splitting depends on chosen enumerations of the fibers; it is not asserted to be canonical.

Proof. A type-preserving correlation preserves P, since old and new types are disjoint. If its restriction is h, incidence preservation says that it maps the fiber over (j, S) bijectively to that over (j, hS). Thus $h \in H$. When $h = 1$, it preserves each fiber. Conversely arbitrary independent permutations of the fibers preserve old–new incidence and all new–new incidences. This proves precisely the asserted kernel, not merely a subgroup contained in it.

For each nonempty fiber choose an enumeration $e_{j,S} : \{1, \dots, m_j(S)\} \to s_j^{-1}(S)$. Extend $h \in H$ by

$$e_{j,S}(a) \longmapsto e_{j,hS}(a).$$

Equality of multiplicities makes this well-defined. It preserves old–new incidence because $p \in S$ if and only if $h(p) \in hS$; the other incidences are preserved by construction. Its inverse is the extension of h^{-1}. Applying successive extensions keeps the index a unchanged, so these extensions form a homomorphism and split restriction. Finally a symmetric group is trivial exactly when its set has at most one element. □

For full correlations one must separately recognize the old object set and allow any actual permutation of new types. Thus a signature that includes a new type does not establish pointwise rigidity if a correlation fixing old objects could still permute those types. The finite examples in the second application verify this distinction explicitly.

Corollary 9 (Recognition by unique shadows)**.** *Suppose every correlation of an extension Y recognizes an old structure X, and its restriction preserves all reconstructed old incidences and colors. Suppose every new object is uniquely determined by its complete old shadow, even across any new types that correlations may permute. Let $G \leq C(X)$ be exactly the old actions whose forced permutations of new objects preserve the prescribed incidences and induce a permutation of the full palette. Then restriction identifies $C(Y)$ with G; $K(Y)$ corresponds to the kernel of the resulting palette action.*

Proof. Unique shadows give pointwise rigidity, and the definition of G gives exactly the permissible extensions. Theorem 1 applies. □

Proposition 10 (Keeping ordered positions)**.** *Let P be a typed incidence system and T a set of distinct ordered d-tuples of old objects, with a prescribed type for every tuple. Form a new system with a center for each tuple t and a port (t, i) for each position $1 \leq i \leq d$. Give ports distinct position types, disjoint from the old and center types, and join (t, i) just to its center and the old object t_i. Keep the old incidences. In the subgroup preserving the old types, center types and each position type, restriction is an isomorphism onto the old automorphisms preserving the tuple sets by type.*

Proof. For a correlation in the stated subgroup, the unique center and old neighbor of a port show that the center image has entries $h(t_i)$ in all positions. Since tuples are distinct, its image and all its ports are forced. Conversely an old automorphism preserving the tuple sets extends by $t \mapsto (h(t_1), \dots, h(t_d))$ and $(t, i) \mapsto (h(t), i)$. Both maps preserve all declared incidences and are inverse under restriction. Apply Theorem 1. □

This is a precise repair of loss of order at the level of a specified role-preserving subgroup. It does not assert that the entire correlation group recognizes those roles, that projective dualities extend, or that the resulting incidence system is a geometry. Those are additional requirements, not consequences of this proposition.

Theorem 11 (Rigid arms encode ordered attachments)**.** *Let X be a finite properly colored graph and let T be a family of pairwise distinct ordered d-tuples of its vertices, with $d \geq 3$ and with a color $c(t)$ assigned to each tuple. Add a center for each tuple and join it to its ith entry by a fresh path of length $\ell_i \geq 2$, where the ℓ_i are pairwise different. Give centers their prescribed colors and give internal path vertices separate colors indexed by (i, k), their arm and distance from the center. All new colors are disjoint from the old colors. No other new incidences are added. Suppose every correlation of the resulting graph Y preserves the old vertex set $V(X)$. Then restriction identifies $C(Y)$ with the subgroup of $C(X)$ preserving T and its partition into tuple colors. Under this identification $K(Y)$ is the kernel of the action on the old colors and tuple colors; all internal path colors are fixed automatically.*

Proof. Each component after deleting the old vertices is a subdivided star with one vertex of degree d, its center. Its arm lengths are $\ell_1 - 1, \dots, \ell_d - 1$ and are pairwise different. Every isomorphism between two such components sends the center to the center, preserves arm indices by their lengths and preserves each distance from the center. Looking also at the unique old endpoint of each arm therefore recovers the entire ordered tuple, not merely its unordered boundary. Distinct tuples have distinct ordered boundaries, so a correlation fixing all old vertices fixes each component pointwise.

A correlation restricts to $C(X)$ because all old incidences and colors are retained. Its action sends each tuple to the ordered tuple of its endpoint images, with one common permutation of the tuple colors. Conversely an old correlation satisfying these conditions extends by that rule on centers and by matching indices and distances on arms. This is an incidence-preserving bijection, and the inverse is obtained from the inverse old correlation. The internal colors are fixed and the other colors transform just as asserted. Apply Theorem 1. □

6 Free-basis incidence systems

The following argument reuses the complete free-basis proof from the earlier phase197 manuscript; its reduced-word and tree arguments are included here in full. Classical reduced-word normal forms and the universal property of free groups remain explicit inputs. No historical priority is asserted. The object set may now be infinite, but the type set remains finite.

Let I be a finite set of cardinality $r \geq 2$, let $F = F(s_i : i \in I)$ be the free group on this basis, and define

$$F_i = \langle s_j : j \neq i \rangle, \qquad F_J = \bigcap_{j \in J} F_j.$$

The objects of type i are the right cosets $F_i g$. Objects of distinct types are incident when the cosets intersect. Denote this incidence system by Γ_I. Use the left permutation convention for the natural action:

$$R_g(F_i h) = F_i h g^{-1}. \tag{F1}$$

Thus $g \mapsto R_g$ is a homomorphism.

7 Free-basis cosets and the Helly condition

A standard free factor generated by a subset of the basis consists exactly of the elements whose reduced words use only that subset. In particular

$$F_J = \langle s_k : k \in I \setminus J \rangle. \tag{F2}$$

This includes $F_\varnothing = F$ and $F_I = 1$.

For a reduced word w and $i \in I$, let $P_i(w)$ be the prefix ending at its last occurrence of s_i or s_i^{-1}; set $P_i(w)$ to the empty word if there is no such occurrence.

Lemma R2.Prefix. For $i \notin J$,

$$w \in F_J F_i \quad \Longleftrightarrow \quad P_i(w) \text{ uses no letter indexed by } J. \tag{F3}$$

Proof. If $w = ab$ with $a \in F_J$ and $b \in F_i$, reduce the concatenation of their reduced words. Cancellation takes place across their common boundary. The remaining word is a prefix of the reduced word of a followed by a suffix of that of b, with no further cancellation. The second part has no occurrence indexed by i, so the prefix ending at the last such occurrence lies entirely in the first part. It avoids J. This also covers complete cancellation of the first part, when the prefix is empty. Conversely, if the specified prefix avoids J, split w immediately after its last i-occurrence. The prefix belongs to F_J, and the suffix has no i-occurrence and belongs to F_i. □

Corollary R2.Product. For every $J \subseteq I$ and $i \notin J$,

$$F_J F_i = \bigcap_{j \in J} F_j F_i. \tag{F4}$$

Proof. By (F3), membership in the right side says that $P_i(w)$ avoids each $j \in J$, which says exactly that it avoids J. The empty intersection causes no exception. □

We prove flag-transitivity directly, so no outside coset criterion is a premise here. Consider the Cayley tree with edges joining g to $s_i^{\pm 1} g$. Every right coset of a standard free factor is a convex subtree, because the unique reduced path between its vertices uses only its allowed generators. A finite family of pairwise intersecting subtrees of a tree has a common vertex. For example, root the tree and select, among the nearest-to-root vertices of the subtrees, one at maximal depth. For any other subtree, take a point in its intersection with the selected subtree. Both nearest vertices lie on the root path to that point; the deeper one lies in both subtrees. It therefore lies in every member of the family. This

proves that every flag of Γ_I consists of cosets with a common representative. Right translation sends that representative to the identity and the flag to the standard flag of its type. In particular every flag extends to a chamber.

Theorem R2.Geometry. The incidence system Γ_I is a flag-transitive, residually connected geometry. The natural group $R(F)$ acts regularly on chambers.

Proof. The common-representative argument proves geometry and flag-transitivity. The stabilizer of the standard chamber is $\bigcap_i F_i = 1$ by (F2). Thus the chamber action is regular and the natural action is faithful.

Translate an arbitrary flag to the standard flag of some type J. Its chambers are uniquely indexed by F_J, and its remaining objects identify with the cosets of

$$F_J \cap F_i = F_{J\cup\{i\}}, \qquad i \notin J,$$

inside F_J. To verify this identification, intersect each incident coset with F_J; it is nonempty by the common-representative property and is a coset of the displayed intersection subgroup. Distinct right cosets remain distinct after nonempty intersection.

The residue is consequently the same free-basis construction on the remaining basis $\{s_i : i \notin J\}$. If at least two types remain, its chamber graph is connected: chambers labeled h and $s_i^m h$ share all objects except possibly type i, and the remaining basis elements generate F_J. Each chamber is a clique with at least two vertices, and consecutive chambers have a common object. The union of this connected family of cliques is the entire residue incidence graph. Thus every residue of rank at least two is connected. □

This proof also yields exactly the generation condition quoted as Theorem 7.4 in the source:

$$F_J = \left\langle F_{J\cup\{i\}} : i \in I \setminus J \right\rangle \quad \text{when } |I \setminus J| \geq 2. \tag{F5}$$

Each remaining basis letter belongs to one of those factors by choosing a different remaining index.

Proof correction. Membership in F_jF_i does not imply that the entire reduced word avoids s_j. Its second factor is allowed to contain that letter. Theorem 7.3's printed proof uses this implication. Formula (F3), rather than avoidance by the whole word, supplies the valid common argument.

8 Palette actions of a free basis

Let $C = C(\Gamma_I)$ and $K = K(\Gamma_I)$ denote the full correlation and type-preserving groups. An automorphism α of F satisfying $\alpha(F_i) = F_{\sigma(i)}$ induces the correlation

$$F_i g \longmapsto F_{\sigma(i)}\alpha(g). \tag{F6}$$

The formula is well-defined because equality of cosets is preserved by α, and incidence is preserved because α preserves nonempty intersection.

Theorem R2.TypeImage. The action of the full correlation group on the palette has image $\mathrm{Sym}(I)$. Thus

$$C/K \cong \mathrm{Sym}(I). \tag{F7}$$

Proof. Every permutation σ of the free basis extends to an automorphism of F by the universal property of the free group, and it sends F_i to $F_{\sigma(i)}$. Formula (F6) therefore realizes every palette permutation. The kernel of the palette action is precisely K by definition. □

There is no claim here that (F7) determines K or the entire group C. The natural subgroup $R(F)$ need not be the full type-preserving group.

The inherited Lean file `BasisPermutation.lean` verifies the generic basis-permutation automorphism, its faithful dependence on the permutation, and the equation transporting the subgroup closure generated by every complementary basis. This is the formal group-theoretic step in the proof of (F7). The coset quotient construction and the inference of the full type quotient are written proofs, not part of that file's formal scope.

9 The normalizer of the chamber-regular group

Write

$$W_I = C_2^I \rtimes \mathrm{Sym}(I)$$

for the signed basis permutations. Each maps s_i to $s_{\sigma(i)}^{\epsilon_i}$ with $\epsilon_i \in \{1, -1\}$ and acts by (F6).

Theorem R2.Normalizer. The normalizer of the natural chamber-regular group is

$$N_C(R(F)) = R(F) \rtimes W_I. \tag{F8}$$

The signed basis subgroup is precisely the stabilizer of the standard chamber inside this normalizer.

Proof. If c normalizes $R(F)$, compose c with the unique natural translation taking its image of the standard chamber back to that chamber. It suffices to consider a normalizing correlation d fixing the standard chamber. Conjugation by d induces an automorphism α of F, since $R : F \to R(F)$ is an isomorphism. Let σ be the palette permutation of d.

The pointwise stabilizer in $R(F)$ of the standard chamber with type i deleted is

$$R\left(\bigcap_{j \neq i} F_j\right) = R(\langle s_i \rangle).$$

Since d fixes the chamber as a set and permutes its types by σ, it sends this stabilizer to $R(\langle s_{\sigma(i)} \rangle)$. Thus α maps each infinite cyclic subgroup $\langle s_i \rangle$ onto $\langle s_{\sigma(i)} \rangle$. An isomorphism of infinite cyclic groups sends a generator to a generator, so $\alpha(s_i) = s_{\sigma(i)}^{\pm 1}$. Hence $\alpha \in W_I$.

A normalizing correlation fixing the standard chamber is determined by α. Indeed every chamber is its unique translate by an element of $R(F)$, so its image is forced by conjugation; every object belongs to a chamber. Conversely every signed basis permutation acts by (F6), fixes the standard chamber, and normalizes the natural group. The chamber-regular subgroup has trivial intersection with that stabilizer, and every normalizer element has the unique factorization already described. This proves (F8). □

In particular, the geometry gives a weak representation of the specified pair

$$(F,\ F \rtimes W_I)$$

when the inner group is written first as in the source's pair notation. This is a positive statement about compatible injections into (K, C); it is not a representation of the full abstract group F in the source's special sense.

Corollary R2.NaturalObstruction. For $r \geq 2$, there is no injective homomorphism

$$\mathrm{Aut}\, F \longrightarrow C(\Gamma_I)$$

whose restriction to $\mathrm{Inn}\, F \cong F$ is the natural embedding R.

Proof. The inner subgroup is normal, so the image of such a homomorphism would lie in $N_C(R(F))$. Injectivity, and equality of the image of the inner subgroup with $R(F)$, would induce an injection

$$\mathrm{Out}\, F \hookrightarrow N_C(R(F))/R(F) \cong W_I.$$

The right side is finite. The left side is infinite: for distinct basis elements s_1, s_2, the automorphisms $s_1 \mapsto s_1 s_2^m$, fixing the other basis elements, induce distinct transformations of the abelianization $\mathbb{Z}^r$, whereas inner automorphisms act trivially there. This is a contradiction. □

The corollary fixes the natural embedding used in the construction. It does not rule out every conceivable weak embedding with a different image of $\mathrm{Inn}\, F$. The source's finite-rank definition separately rules out an *exact* representation of a nonabelian free group: its outer quotient would have to embed into the finite palette permutation group. That observation is already made in [L126, §7].

10 Weak representations from panel permutations

The natural-normalizer obstruction fixes the prescribed chamber-regular inner action. It does not obstruct other compatible embeddings in the source's definition of a weak representation. The following construction settles that distinction without changing the incidence geometry.

Theorem 12 (An unrestricted panel-permutation subgroup)**.** *For a finite free basis of size $r \geq 2$, the type-preserving group of Γ_I contains* $\mathrm{Sym}(\mathbb{Z} \setminus \{0\})$*. Consequently it contains every countable group.*

Proof. Fix $i \in I$. Write elements of F in their unique reduced syllable form $s_{j_1}^{n_1} \cdots s_{j_k}^{n_k}$, with nonzero integer exponents and consecutive indices different. Extend any $\pi \in \mathrm{Sym}(\mathbb{Z} \setminus \{0\})$ by $\pi(0) = 0$. Define $\Phi_\pi : F \to F$ as follows: if the final syllable is s_i^n, replace it by $s_i^{\pi(n)}$; otherwise leave the word unchanged. The empty word is fixed. Nonzero exponents remain nonzero, so reduced syllable forms stay reduced. It follows directly that $\Phi_\pi \Phi_\rho = \Phi_{\pi\rho}$ and $\Phi_\pi^{-1} = \Phi_{\pi^{-1}}$.

For $j \in I$, call g, h j-adjacent when $h = s_j^m g$ for some integer m; these are precisely the chambers in one j-panel. The map Φ_π preserves this relation in both directions. On the panel $\langle s_i \rangle$ this is immediate: it is an arbitrary permutation of that panel fixing its identity chamber. Outside that case, multiplication on the left changes or cancels only the first syllable. It leaves the final i-syllable unchanged, if one is present; even when a two-syllable word loses its first syllable, both orders of the two operations give $s_i^{\pi(n)}$. If no final i-syllable is present the words remain fixed, except for the already handled panel at the identity. Thus adjacent images are again in the same j-panel. Applying the inverse proves the converse.

The chambers in an object $F_j g$ are exactly those reachable from g using panels of labels other than j, because F_j is generated by the corresponding cyclic factors. Therefore Φ_π permutes the right cosets of every F_j. Define its object action by $F_j g \mapsto F_j \Phi_\pi(g)$. The preceding observation proves well-definedness and bijectivity. Two objects are incident exactly when their sets of chambers intersect, so this action preserves incidence and types. Distinct permutations π give distinct object actions: the type i cosets $F_i s_i^n$ are distinct, since $F_i \cap \langle s_i \rangle = 1$, and their images detect $\pi(n)$. This proves the symmetric-group embedding.

Finally a countable group embeds faithfully in the symmetric group of its underlying set by its regular action. Identify that set with a subset of $\mathbb{Z} \setminus \{0\}$ and fix the complement to obtain the asserted embedding. □

Corollary 13 (Weak representation of a finite-rank free group)**.** *The geometry Γ_I is a weak incidence representation of the abstract free group F in the source's precise sense. Both compatible images may be taken inside its type-preserving group.*

Proof. An automorphism of F is determined by the images of its finitely many basis elements. The set F^r is countable, so $\mathrm{Aut}(F)$ is countable. Embed $\mathrm{Aut}(F)$ in $K(\Gamma_I)$ using Theorem 12, and restrict this injection to $\mathrm{Inn}(F)$. Composing the first map with $K(\Gamma_I) \hookrightarrow C(\Gamma_I)$ gives the required commuting pair of injections. □

This is an existence construction with a different inner image. It neither extends the natural chamber-regular action nor contradicts Theorem R2.Normalizer and its natural-embedding obstruction. No quotient identification is inferred from these weak injections.

11 Chamber stabilizers and the splitting issue

A geometry is *thin* when every panel belongs to exactly two chambers. The chamber graph joins chambers differing in one object. This graph need not be connected merely from the definition that every flag extends to a chamber.

Proposition 14 (The connected case)**.** *Suppose a thin geometry has a connected chamber graph and its type-preserving automorphism group K is chamber-transitive. Write C for its full correlation group. Then K acts regularly on chambers. For any chamber D, its setwise stabilizer C_D is a complement to K in C:*

$$C = K \rtimes C_D, \qquad C_D \cong C/K.$$

Proof. An element of K fixing D fixes its objects individually. It fixes each adjacent chamber: the common panel is fixed, and of its two chambers one is already fixed, so the other is fixed as well. Connectedness propagates this to every chamber. Every object lies in a chamber; hence the element is the identity. Chamber transitivity now gives regularity. Given $c \in C$, choose $k \in K$ with $kD = cD$. Then $k^{-1}c \in C_D$. Thus $C = KC_D$, while $K \cap C_D = 1$. Normality of the palette kernel K proves the semidirect product and the quotient identification. □

Proposition 15 (Thin doubling of a bipartite graph)**.** *Let $B = (P \sqcup Q, E)$ be a finite connected simple bipartite graph with $|P| = |Q| \neq |E|$. Assume that its part-preserving automorphism group H^+ is transitive on E and that some automorphism exchanges its parts. Write $H = \mathrm{Aut}(B)$, so $H/H^+ \cong C_2$. There is a finite thin, flag-transitive rank-three geometry $\Delta(B)$ with*

$$C(\Delta(B)) \cong C_2^{P \sqcup Q \sqcup E} \rtimes H, \qquad K(\Delta(B)) \cong C_2^{P \sqcup Q \sqcup E} \rtimes H^+.$$

If the exchanging coset $H \setminus H^+$ contains no involution, the first group does not split over the second. Its chamber graph has one component for each edge of B.

Proof. For each $v \in P \sqcup Q \sqcup E$ take two objects $(v, 0), (v, 1)$, with the three displayed sets as their types. Incidence ignores the second coordinate: a P-object and a Q-object are incident precisely when their underlying vertices form an edge of B; an edge-object is incident with precisely the objects over its two endpoints. Distinct objects of one type are nonincident; we use the source's irreflexive incidence convention. A chamber is exactly an edge together with independent choices of one of the two objects over each endpoint and that edge. Every incident pair extends to such a chamber, as does every single object since the graph is connected and has an edge. Thus every flag extends to a chamber. For a panel, the underlying missing vertex or edge is unique, and its two copies give exactly two completions. This proves thinness.

The independent flips of all pairs form $N = C_2^{P \sqcup Q \sqcup E}$. The group H lifts by leaving second coordinates unchanged. The subgroup $N \rtimes H^+$ is transitive on chambers, using edge transitivity followed by the three endpoint/edge flips. It is therefore transitive on flags of each type.

Every correlation fixes the edge type: its size $2|E|$ differs from both other type sizes. It either fixes or exchanges the other two types. Within each type the two copies of an underlying object are exactly the classes with equal neighborhoods in the other types. For distinct vertices of P or Q, their sets of incident edge-objects are disjoint and nonempty, so their neighborhoods differ. Distinct edge-objects have distinct pairs of endpoints because B is simple. Consequently these pairs are intrinsically determined, and every correlation descends to an automorphism of B, possibly exchanging its parts. Its action on edges is forced by its action on vertices. The restriction kernel consists exactly of the independent flips N. The explicit lifts split this restriction and prove both group formulas.

The quotient by the type-preserving subgroup is C_2. A complement would contain an involution exchanging the two vertex types. Its projection to H would be an involution in $H \setminus H^+$, contrary to the stated hypothesis. Finally adjacent chambers have the same underlying edge; the eight chambers over each edge form the connected three-dimensional cube under changes of a single choice. These are precisely the chamber components. □

The second application supplies a finite graph satisfying the nonsplitting hypothesis, with a complete elementary certificate of its full automorphism group. Thus the unqualified source observation is false, whereas the connected-chamber statement above holds by its direct proof. The following verification discussion concerns the two inherited programs and restriction module used for the shared

finite constructions and first application. The second application's separate scope record also includes the full optional finite/free-word checks, the required graph certificate with its complete data, and the inherited basis-permutation and shadow-kernel formal declarations. The certificate is needed only for the specific nonsplit graph, not for the generic doubling argument.

12 Attribution and verification

The chamber, cleaning and ordered-pair constructions, their symmetry assertions and the finite normal-pair theorem are due to Díaz Ramos, Molinier and Viruel. The restriction-and-extension criterion is elementary classical reasoning; its explicit separation from recognition and rigidity is an organization of the proofs, not a priority claim. Lemma 2 isolates the component argument also used in the recursive application. No external graph representation theorem is needed to make the proofs above complete.

The accompanying `RecognizableRestriction.lean` constructs the restriction homomorphism from its actual action on an embedded vertex set, proves its kernel description, derives injectivity from pointwise rigidity and surjectivity from existence of extensions, constructs the group equivalence, and transports a distinguished subgroup. It does not formalize the incidence or degree arguments proving those hypotheses for any gadget. The complete chamber, cleaning, ordered-pair and component arguments above remain written proofs.

The two complete GAP programs in the separate appendix provide bounded checks, including exceptional graphs and small normal pairs. Their computations do not replace any quantified theorem above. GAP and GRAPE's exact finite automorphism algorithms remain part of the trust boundary of those checks; Lean uses its kernel and the pinned mathlib infrastructure. Neither the number of programs nor the number of formal declarations measures mathematical compression.

References

[1] A. Díaz Ramos, R. Molinier and A. Viruel, *Every finite group is represented by a finite incidence geometry*, arXiv:2511.02410v2, 16 December 2025.